

Three question features: what students see, what the database needs

Question Bank, PYQ Mix and Past Papers all draw from the same 18,651 questions. What separates them is how much time a student has.

The three, at a glance

FEATURE	SESSION	UNIT	CLOCK	FEEDBACK
Question Bank	2 min	One question	None	Instant, per question
PYQ Mix	20 min	A set of 20	Yes	Score at the end
Past Papers	3 hours	A whole real paper	Yes, sectional	Score + section breakdown

A student picks by the time they have, not by the content — the questions are the same in all three. **AI-Curated Exams is not a fourth feature**; it is PYQ Mix with the questions chosen by weakness instead of at random. One toggle, not one more menu item.

Shared foundation: five tables

Everything below sits on these. Only two of them are about content; the other three record what a student did.

papers	one row per source document
paper_id	primary key – which document this is
exam_key	which real exam it recalls (many papers → one exam)
bank_role	SBI / IBPS · PO / Clerk / RRB
exam_type_year	Prelims / Mains · 2019
shift	often null
duration_min	total time, for mocks
total_marks	for scoring
is_canonical	show this one, hide the near-duplicate recalls
source_pdf	provenance
directions	passages, shared by several questions
paper_id	← part of the key. IDs repeat across papers
direction_id	d001...d030
body	the passage text
questions	
q_id	primary key
paper_id	which document it came from
q_num	position in that paper
section	Reasoning / Quantitative / English / GA
topic	finer than section
difficulty	1–5
direction_id	nullable, resolved with paper_id
stem	the question
options	{ a...e }
answer	'a'...'e'
explanation	why
marks	usually 1
negative_marks	usually 0.25 – banking exams penalise wrong answers
content_hash	same question from two sources = one row
is_active	retire a bad question without deleting it
attempts	one row per sitting
attempt_id	user_id mode('bank' 'mix' 'paper')
paper_id	null unless mode='paper'
started_at	submitted_at score
attempt_answers	one row per question answered
attempt_id	q_id chosen is_correct time_ms

Two things worth calling out. **directions** is its own table because the passage text is currently copied onto every question that shares it — and passage IDs repeat across papers, so the key must be (paper_id, direction_id) . And **attempt_answers** is what makes **mastery possible**; without it the grade rail has nothing to measure.

WHAT THE STUDENT SEES

Question Bank 12 / 240 solved
SBI · PO · Reasoning

Q. A, B, C, D and E sit in a row...

- ☐ 3rd to the right
- ☒ 5th to the left
- ☐ 2nd to the left
- ☐ 3rd to the left

[Check] Skip →

- One question at a time. Answer, told right or wrong, next.
- **No clock, no submission.** Close the app mid-question and nothing is lost.
- The counter is scoped to the filter — 12 / 240 , not 12 / 18,651 . A finishable number, not a demoralising one.
- Filters live in the address, so a link can be shared.

SCHEMA IT NEEDS

TABLE	FIELDS USED	STATUS
questions	stem, options, section, is_active	HAVE
questions	answer, explanation	0% FILLED
papers	bank, role, exam_type, year	93–97%
directions	body — for comprehension sets	HAVE
attempt_answers	chosen, is_correct — powers the counter	TO BUILD

HOW QUESTIONS ARE SELECTED

```
SELECT q.* FROM questions q
JOIN papers p USING (paper_id)
WHERE q.is_active
      AND p.bank = $1 AND p.year = $2 AND q.section = $3
      AND q.q_id NOT IN (already answered by this user)
```

```
ORDER BY q.q_num  
LIMIT 1 OFFSET $n
```

One at a time, skipping what they've already done. No timer means no `attempts` row — just an `attempt_answers` entry per question, which is what feeds mastery later.

WHAT THE STUDENT SEES

Build a set

How many [20 ▼]

Section [Reasoning ▼]

Years [Any ▼]

Source (•) Mixed from past years
() Target my weak areas

[Start]

then: Q 7 of 20 ⌚ 13:42 left

- Set it up once, then a timed run with a progress bar.
- **It ends.** That's the point — a finishable task before bed is what builds the daily habit.
- Score and a per-section breakdown at the end.
- “Target my weak areas” is the AI mode. It needs `topic` data we don't have yet.

SCHEMA IT NEEDS — EVERYTHING THE BANK USES, PLUS

TABLE	FIELDS USED	STATUS
attempts	started_at, submitted_at, score, mode='mix'	TO BUILD
questions	marks, negative_marks — for scoring	ADD
questions	difficulty — to balance a set	ADD
questions	topic — for the weak-areas mode	0% FILLED
attempt_answers	time_ms — per-question pacing	TO BUILD

HOW QUESTIONS ARE SELECTED

```

-- Mixed: spread across years, balanced by difficulty
SELECT q.* FROM questions q
JOIN papers p USING (paper_id)
WHERE q.is_active AND q.answer IS NOT NULL
      AND q.section = $1 AND p.bank = $2
ORDER BY random()
LIMIT 20;

```

```
-- Weak areas: same, but weighted toward topics
-- the student gets wrong most often
ORDER BY (SELECT accuracy FROM user_topic_mastery
          WHERE topic = q.topic AND user_id = $me) ASC,
          random()
LIMIT 20;
```

Note the `answer IS NOT NULL` . A timed set that cannot be scored is worse than no feature — so this screen stays behind the answer-key work.

WHAT THE STUDENT SEES

Past Papers		SBI ▾	P0 ▾
Mains			
2022	190 questions	3 hrs	▶
2019	190 questions	3 hrs	✓ 142
Prelims			
2022	100 questions	1 hr	▶

- Browsed by exam, then stage, then year — the way aspirants already think, and the way Oliveboard and Adda247 arrange it.
- Papers already attempted show the score instead of a play button.
- Attempting one is the real thing: sectional timing, negative marking, score at the end.
- Only one paper per real exam is listed — `is_canonical` hides the near-duplicate recalls of the same exam.

SCHEMA IT NEEDS — THE MOST DEMANDING OF THE THREE

TABLE	FIELDS USED	STATUS
papers	duration_min, total_marks	ADD
papers	exam_key, is_canonical — dedupe recalls	ADD
papers	section_durations — prelims are sectionally timed	ADD
questions	q_num — order must be exact	HAVE
questions	marks, negative_marks	ADD
attempts	mode='paper', paper_id, score	TO BUILD

HOW QUESTIONS ARE SELECTED

```

-- No sampling. The paper is the paper.
SELECT q.*, d.body AS passage
FROM questions q
LEFT JOIN directions d
  ON d.paper_id = q.paper_id      -- both keys, always
 AND d.direction_id = q.direction_id

```



```
WHERE q.paper_id = $1  
ORDER BY q.q_num;
```

The join uses **both** `paper_id` and `direction_id`. Passage IDs are only unique inside a paper — all 30 of ours repeat across papers. Join on `direction_id` alone and students read a passage from the wrong exam.

Extraction: getting questions in

The source is 305 JSON files in the question-bank repo, one per paper, with questions nested inside. Two ways in, one shared function.

BULK — A COMMAND, NOT A SCREEN

```
bun run db:import ../question-bank/India/banking/papers-deduped --dry-run

305 papers · 18,651 questions
  new          305 papers, 18,651 questions
  duplicates    0
  ▲ 419 questions have no options → marked inactive
  ▲ 18,651 questions have no answer → cannot be used in timed modes
```

- **Dry run first, always.** It reports what would change and writes nothing.
- **Safe to re-run.** Keyed on `paper_id`, with `content_hash` per question. Running it twice changes nothing — which matters, because every parser improvement means running it again.
- Uploading 305 files through a browser is the wrong tool; the questions live in git, so the sync belongs in a script.

ONGOING — THE ADMIN SCREEN

- Drop in one or more JSON files.
- Same preview, then confirm.
- Calls the same function as the CLI — one code path, so they can't drift.

WHAT EXTRACTION STILL HAS TO SOLVE

GAP	EFFECT	WHERE IT COMES FROM
answer	0% — blocks PYQ Mix and Past Papers entirely	Parser must extract answer keys, or key them in
explanation	No learning moment	Parser, or generated later
section	15.5% — no section filter	Classifier over the stem
topic	0% — no weak-area mode, no mastery	Classifier
marks	Not in source at all	Set per exam pattern, not per question
exam_key	Duplicate mocks of one exam	Group by bank+role+type+year+shift

Build order

#	STEP	BLOCKED BY
1	Schema — five tables, in Drizzle	Nothing
2	Import script, dry-run first	Nothing
3	Question Bank — browse and filter	Nothing (answers make it useful, not possible)
4	Answer keys into the data	Parser work
5	PYQ Mix — timer, scoring	Step 4 + marks
6	Past Papers — sectional timing	Step 4 + durations
7	Section & topic classification	Unlocks weak-area mode and mastery

Steps 1–3 can start today. Everything from 5 onward waits on answer keys — **a timed test that cannot mark itself is not shippable.**

Counts from every file under `India/banking/papers-deduped/` at commit `0ff28ea` — 305 papers, 18,651 questions.